EKOS · EKOS LAKEHOUSE · IDEMPIERE ON ORACLE

# From a legacy estate nobody fully knows to documented, reviewed, reconciled marts

A compiler for enterprise knowledge read iDempiere's code, documentation and live Oracle catalog; an operator then took it through seven steps in a web console — every number here from a measured run.

THE ESTATE

# iDempiere 12: a real ERP, twenty years old

**903**
tables, 160 views, 62 routines

**3,686**
foreign keys · 12,958 check constraints

**4,820**
Java types · 763 generated model classes

**3,356**
migration scripts → 31,620 schema changes

The usual problem: the knowledge is spread over code, scripts, a docs site and the database, and the people who knew why are gone. Before building a lakehouse on it, someone has to say what it is.

THE APPROACH

# Compile the knowledge, then let a person decide

- **EKOS** observes sources without interpreting them, compiles them through deterministic passes and keeps every conclusion with its evidence in an append-only ledger.
- **No language model reads the sources.** The one model call of the workflow drafts a mart from plain words — and every term it writes is resolved against the approved model, or asked back.
- **Decisions are a person's, and bound to what was shown:** a sign-off carries a digest; when a fact changes, the sign-off turns stale by itself.
- **EKOS Lakehouse** turns the approved model into an Iceberg lakehouse: extraction, dbt silver and gold, Dagster.

THE WORKFLOW

# Seven steps, all from the console

| STEP | THE OPERATOR | ON THIS TESTBED | STATE |
|---|---|---|---|
| 1 Analyse | adds sources, clicks Init & compile | census = compiled (903 / 160 / 62) | done |
| 2 Documentation | Generate, Preview, Publish | 1,060 pages on this repo's wiki | done |
| 3 Review | decides, signs each subject off | 4 of 6 signed; 195 conflicts for a person | partial |
| 4 Requirements | describes marts in words | 4 approved, nothing invented | behind 3 |
| 5 Extraction | reviews the plan per table, approves | 715 incremental · 134 full · 1 refused | behind 3 |
| 6 Bronze + silver | generates, loads | 165-table subset · 3,664 / 3,667 tests | behind 3 |
| 7 Marts | Propose, Merge, Build, Reconcile | 4 marts, every total equal to silver's | behind 3 |

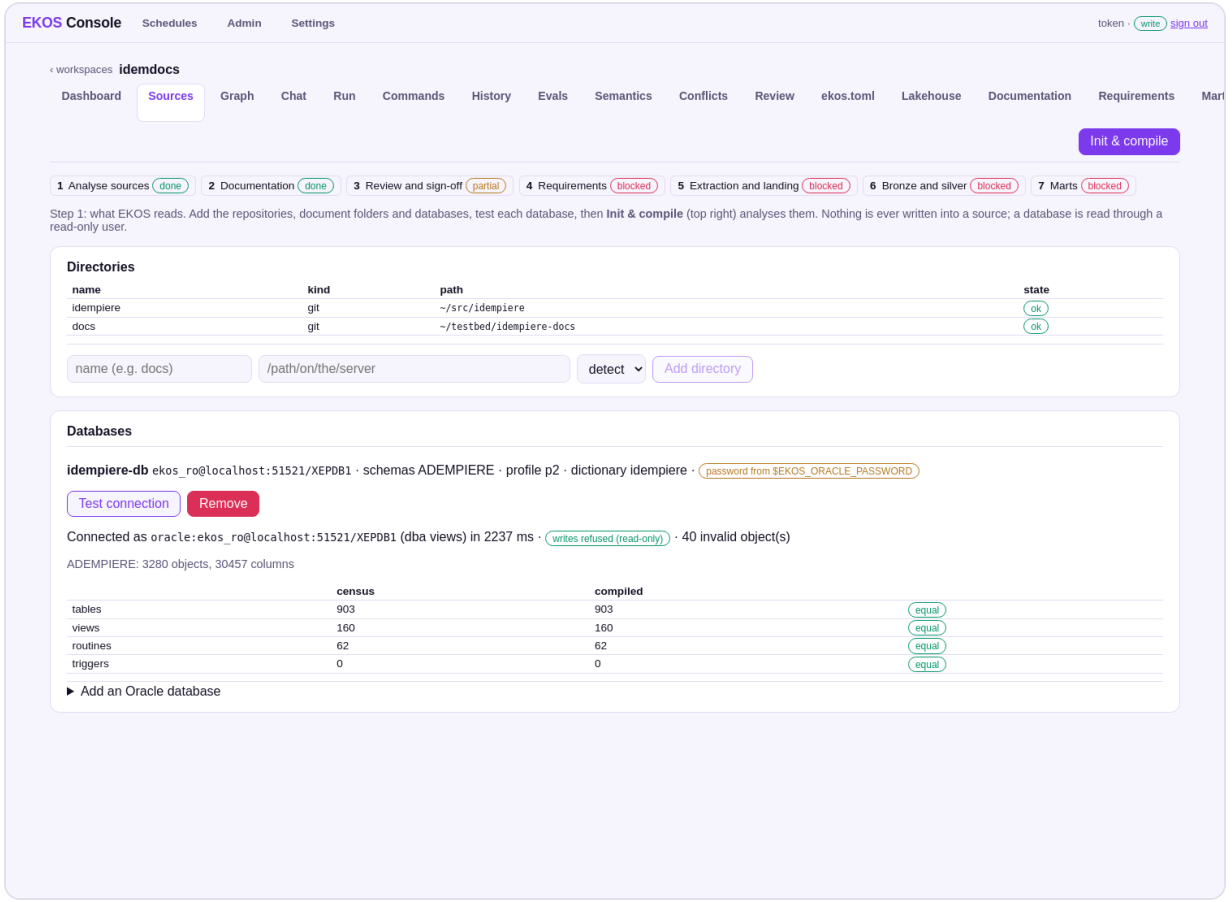
"Behind 3": built and run, shown blocked because step 3 is not fully signed — the gate is the product's, not bypassed.

Model calls in every recorded run of steps 1, 2, 5 and 7: 0

STEP 1 — ANALYSE

# Code, docs and the live catalog in one ledger

- Java declarations and the application model; Oracle SQL and PL/SQL; migration scripts as schema history; docs with their tickets; the live catalog read inside a read-only transaction.
- 446 places where code and database disagree, each with its evidence.
- Per table: how rows are updated and deleted.
- Pipeline: build 16 s · recover 4 min · commit 8 min · **0 model calls**.

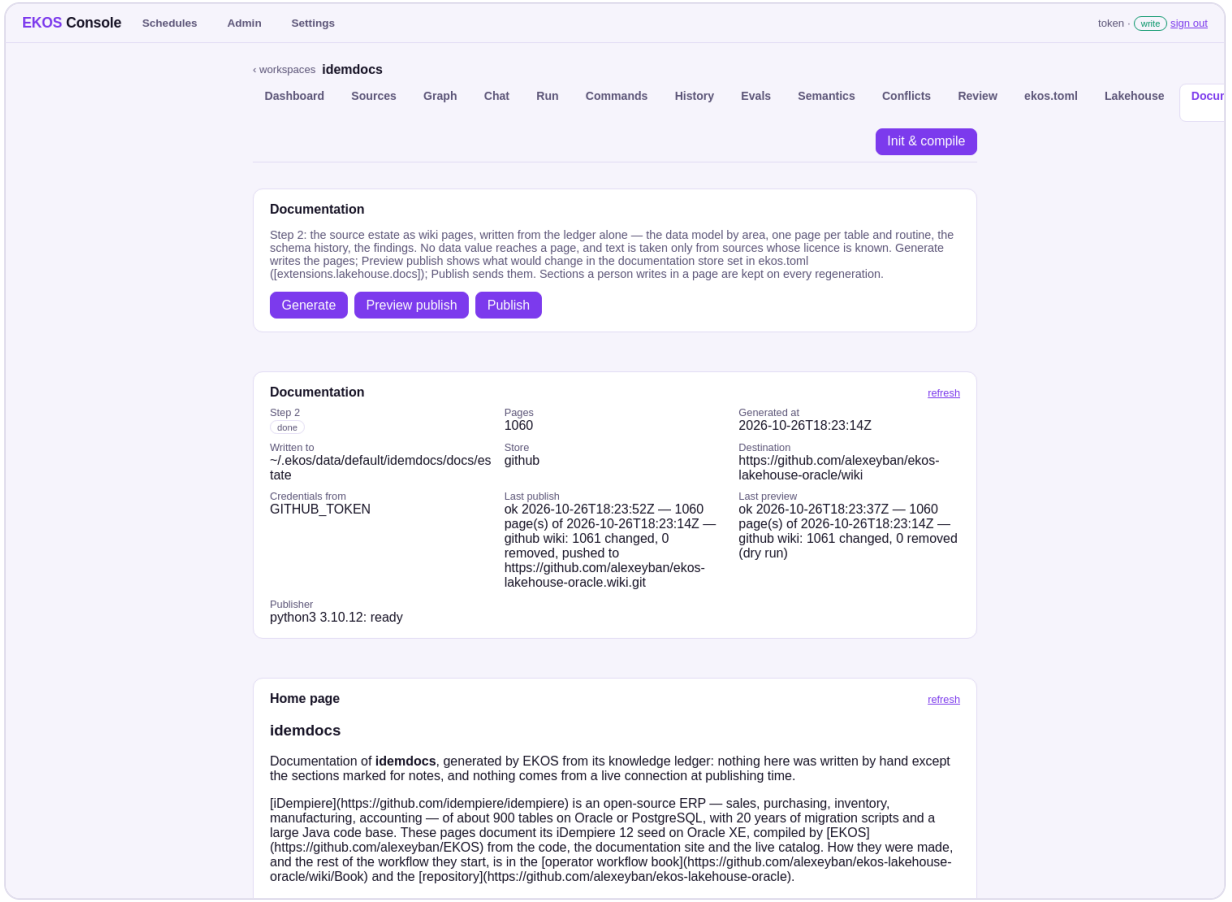



Census reconciled against DBA_OBJECTS: 10,613 objects, 0 mismatches

STEP 2 — DOCUMENTATION

# 1,060 wiki pages, written from the ledger alone

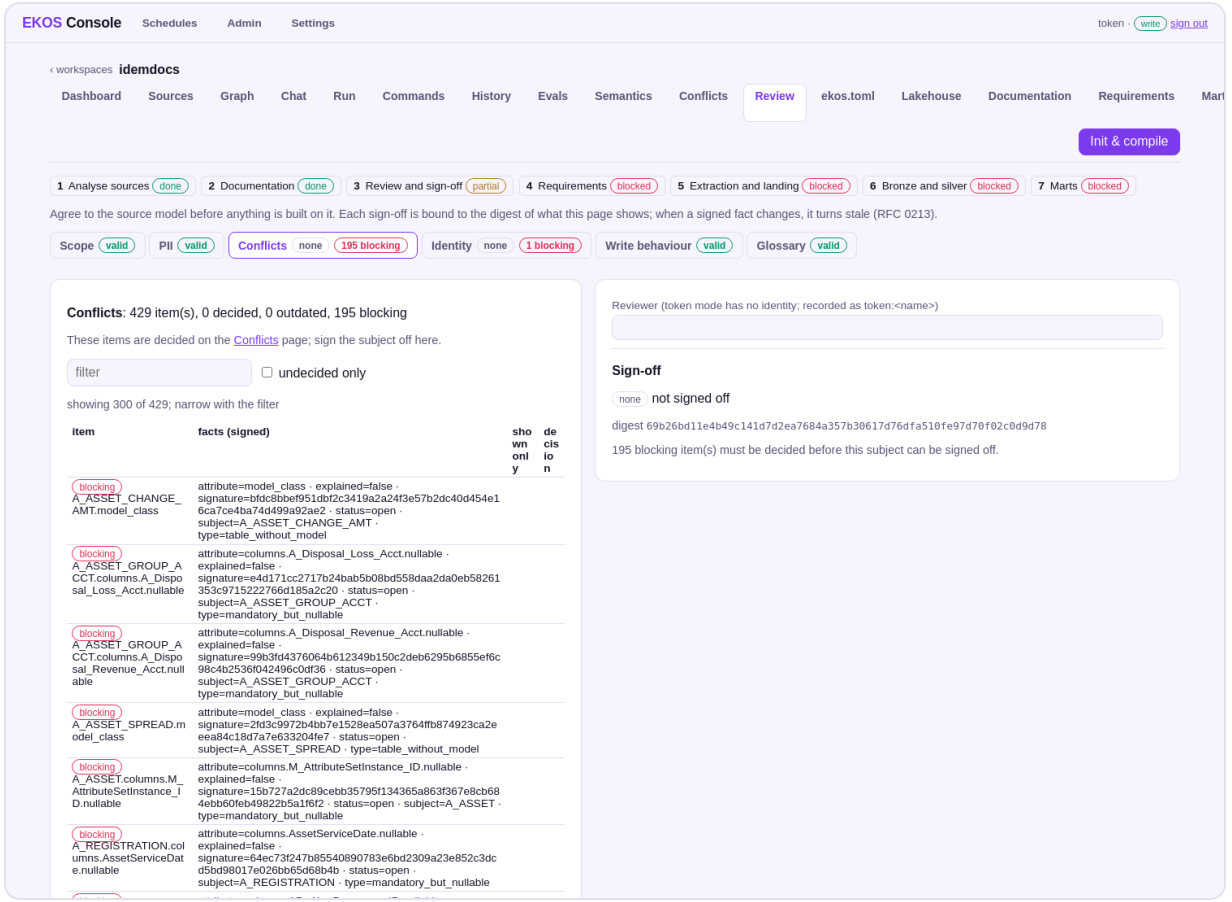
- The data model by area with ER diagrams, a page per table and routine, schema history, findings, a glossary.
- **No data value** reaches a page; text only from sources with a known licence.
- Published to this repository's GitHub wiki from the **Documentation** page — preview first, then publish.



STEP 3 — REVIEW AND SIGN-OFF

# A person signs what they were shown

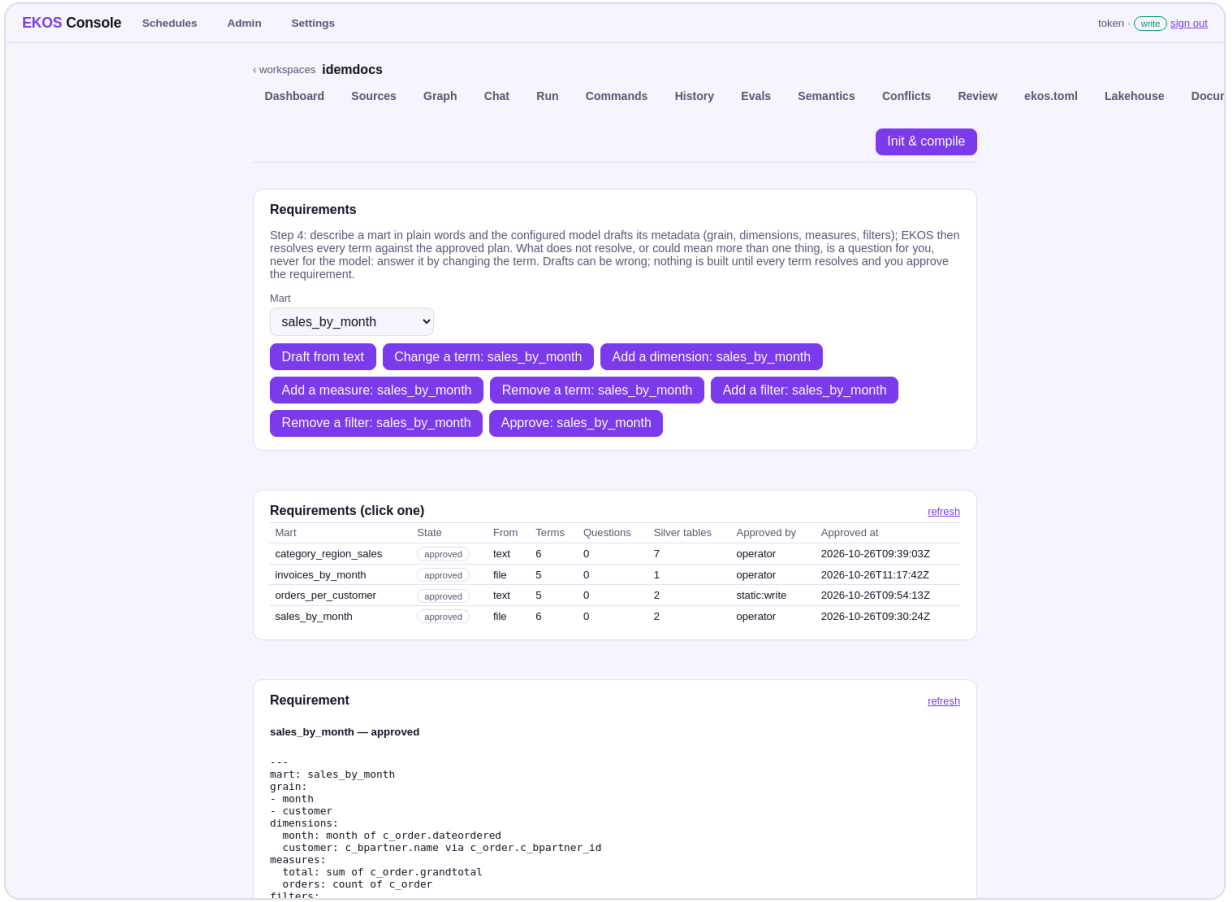
- Six subjects: scope, PII, write behaviour, glossary, conflicts, identity.
- A sign-off is bound to a digest. Adding a column in Oracle turned the PII sign-off stale, naming the column.
- 4 of 6 signed; 195 code ↔ database conflicts wait for a person — the software shows them, it does not decide them.



STEP 4 — REQUIREMENTS

# The one model call, and nothing invented

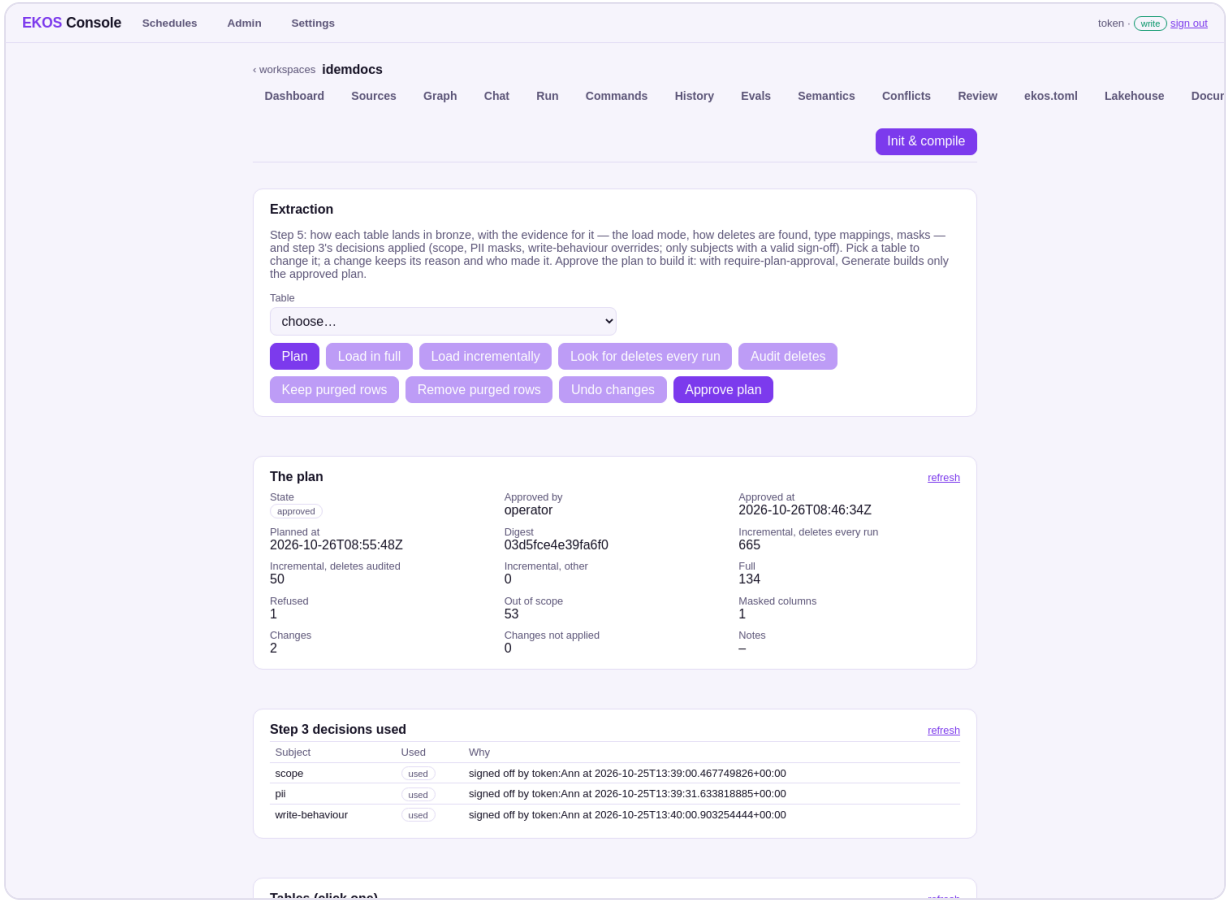
- "Number of completed sales orders per customer and month": local `llama3` drafts the mart's header.
- EKOS resolves every term against the approved model — columns, join paths along declared keys, filters.
- What does not resolve is a question **for the operator**, never fed back to the model.
- Approval records the requirement's digest.



STEP 5 — EXTRACTION

# How each table lands, with the evidence

- Incremental on `UPDATED` where the code keeps it; full where some code skips it.
- Deletes found by a bucketed key diff, with a mass-delete guard; purged rows kept.
- Step 3's scope and PII masks applied; a number no Iceberg type can hold is refused, never rounded.
- The operator changes what they disagree with, with a reason, and approves the plan's digest.



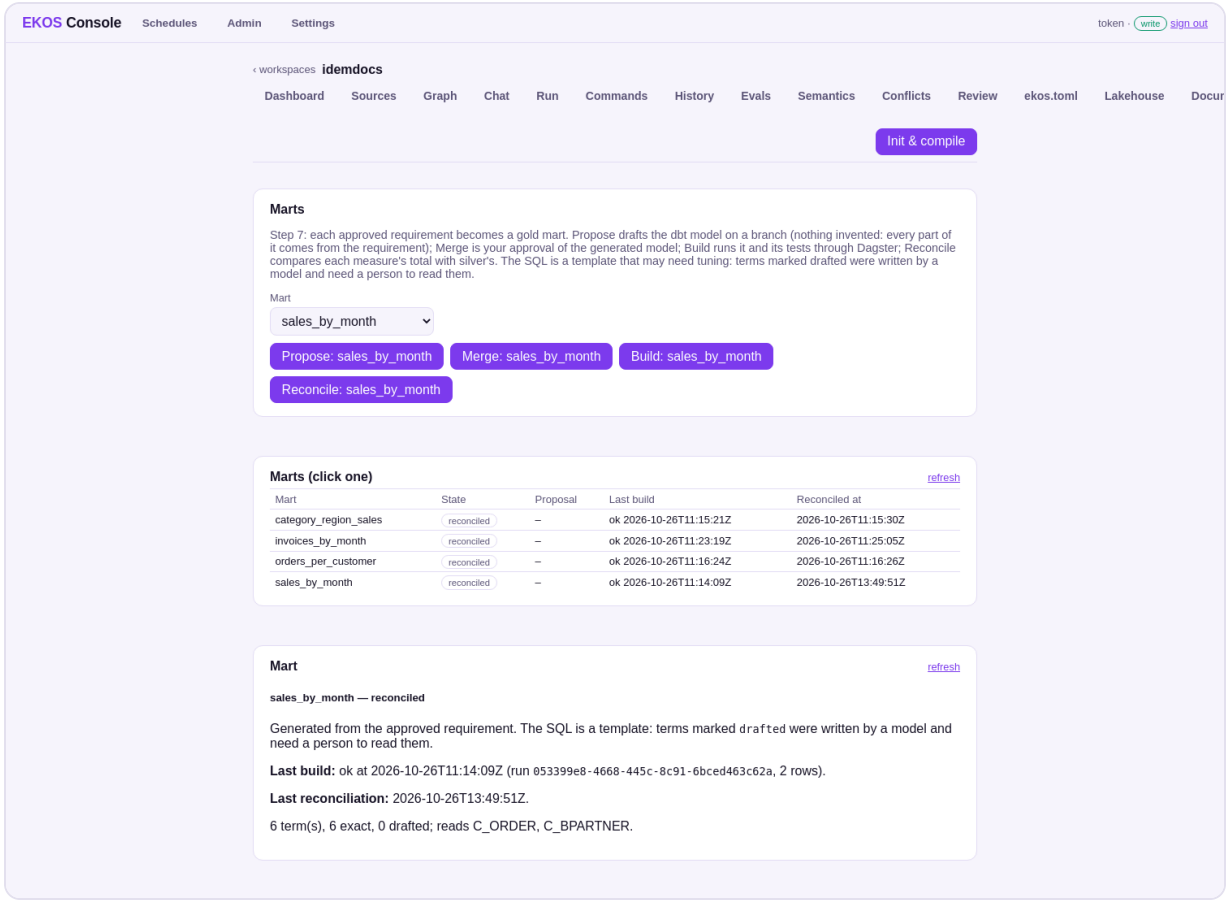

715 incremental · 134 full · 1 refused · 53 out of scope · 1 masked column

STEPS 6 AND 7 — THE LAKEHOUSE

# Silver from the source model, marts reconciled

Generated extraction SQL, dbt-trino silver with tests from the application's own code lists and checks, Dagster. A 165-table subset: **3,664 of 3,667 tests pass**.

| MART · MEASURE | MART | SILVER |
|---|---|---|
| sales_by_month · total | 379.90 | 379.90 |
| sales_by_month · orders | 3 | 3 |
| category_region_sales · sales | 358.40 | 358.40 |
| orders_per_customer · count of orders | 7 | 7 |
| invoices_by_month · invoiced | 541.02 | 541.02 |
| invoices_by_month · invoices | 4 | 4 |



WHAT IS GUARANTEED

# Measured, not promised

- **No row values in the ledger:** 228 text values sampled from 36 business columns — 0 in anything read from the database.
- **No model call outside step 4:** every recorded run log ends with its count — 0 in steps 1, 2, 5 and 7.
- **Every number traceable:** run logs per command; the pipeline is a git repository; approvals are ledger facts.
- **The plan's release gates:** 8 met, 2 partial, 2 blockers — open conflicts for a person, and three delete scenarios still to plant on the testbed.

NOT DONE, HONESTLY

# What this demo does not show

- A complete step-3 sign-off: 195 conflicts and one identity match are a person's work.
- The full 849-table load: a 15 GB laptop ran a 165-table subset.
- Cascade, keyless and permissions-loss deletes on the testbed (designed and classified, not planted).
- CDC with Debezium: optional, built only when an operator asks for it.

EVERYTHING IS HERE

# github.com/alexeyban/ekos-lakehouse-oracle

- **book/** — the seven steps, generated from the console (PDF included)
- **wiki** — the 1,060 documentation pages and the book
- **pipeline/** — the generated bronze SQL, dbt silver and gold, Dagster
- **src/** — EKOS and EKOS Lakehouse at the exact commits · **stack/**, **testbed/** — the containers